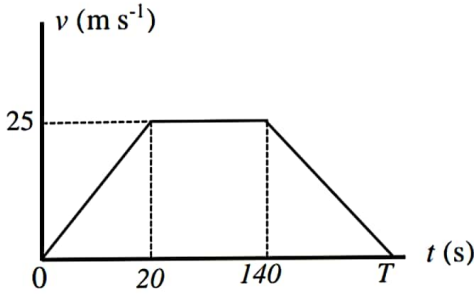


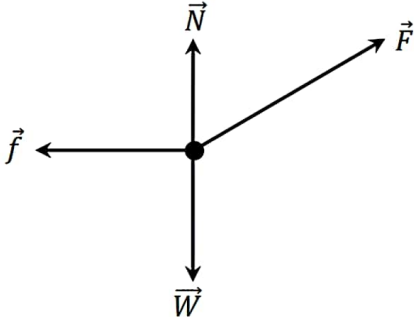


NO.	ANSWER SCHEME												
1(a)	$V = (5.0 \times 10^{-2})^3 = 1.25 \times 10^{-4} \text{ m}^3$ $m = 350 \times 10^{-3} \text{ kg}$ $\rho = \frac{m}{V} = \frac{350 \times 10^{-3}}{1.25 \times 10^{-4}}$ $= 2.8 \times 10^3 \text{ kg m}^{-3}$												
1(b)	<table><tr><th>F</th><th>x – component</th><th>y – component</th></tr><tr><td>F_1</td><td>$= 3.0$</td><td>$= 0$</td></tr><tr><td>F_2</td><td>$= 5.0 \cos 60^\circ$ $= 2.50$</td><td>$= 5.0 \sin 60^\circ$ $= 4.33$</td></tr><tr><td>$\sum F$</td><td>$= 5.50$</td><td>$= 4.33$</td></tr></table> $\sum F = \sqrt{(F_x)^2 + (F_y)^2}$ $= \sqrt{(5.50)^2 + (4.33)^2}$ $= 7.0 \text{ N}$ $\theta = \tan^{-1} \frac{F_y}{F_x} = \frac{4.33}{5.50}$ $= 38.2^\circ \text{ above positive } x \text{ - axis}$	F	x – component	y – component	F_1	$= 3.0$	$= 0$	F_2	$= 5.0 \cos 60^\circ$ $= 2.50$	$= 5.0 \sin 60^\circ$ $= 4.33$	$\sum F$	$= 5.50$	$= 4.33$
F	x – component	y – component											
F_1	$= 3.0$	$= 0$											
F_2	$= 5.0 \cos 60^\circ$ $= 2.50$	$= 5.0 \sin 60^\circ$ $= 4.33$											
$\sum F$	$= 5.50$	$= 4.33$											
TOTAL													



NO.	ANSWER SCHEME
2(a)(i)	 Axes and label : 1 mark Shape : 1 mark
2(a)(ii)	$s = \text{area under } v - t \text{ graph}$ $4000 = \left(\frac{1}{2} \times (25) \times (T + 120) \right)$ $T = 200 \text{ s}$
2(b)(i)	$v = u + at$ $60 = 12 + a(4)$ $a = 12 \text{ m s}^{-2}$
2(b)(ii)	$s = ut + \frac{1}{2}at^2$ $= (12)(4) + \frac{1}{2}(12)(4)^2$ $= 144 \text{ m}$
2(c)	$u = 0$ $v = u + at$ $= 0 + (-9.81)(4.0)$ $= -39.24 \text{ m s}^{-1}$: downward
	TOTAL

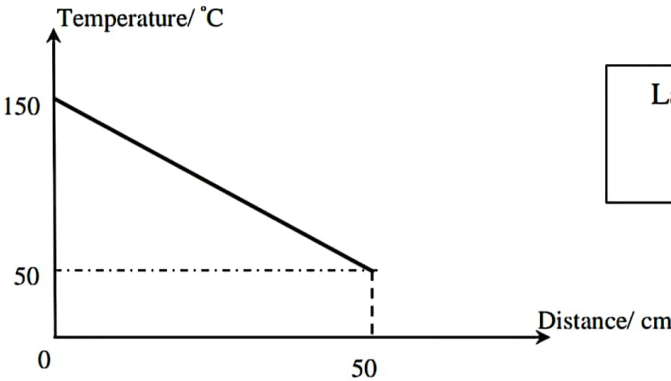
NO.	ANSWER SCHEME
3(a)	$F = -540 \text{ N}$ (Force opposes the motion) $Ft = \Delta P = mv - mu$ $t = \frac{65(0 - 175)}{(-540)} = 21.06 \text{ s}$
3(b)	$\sum p_i = \sum p_f$ $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ $(7.0)(0.60) + (4.5)(0) = (7.0)(0.30) + (4.5)v_2$ $v_2 = 0.47 \text{ m s}^{-1}$
3(c)	<u>x - component</u> $\sum p_i = \sum p_f$ $(55)(65) + (24)(0) = (55)v_x + (24)(85 \cos 20^\circ)$ $v_x = 30.12 \text{ m s}^{-1}$ <u>y - component</u> $\sum p_i = \sum p_f$ $0 = (55)v_y + (24)(85 \sin 20^\circ)$ $v_y = -12.69 \text{ m s}^{-1}$ $v = \sqrt{(v_x)^2 + (v_y)^2}$ $= \sqrt{(30.12)^2 + (-12.69)^2}$ $= 32.68 \text{ m s}^{-1}$ $\theta = \tan^{-1} \frac{v_y}{v_x} = \tan^{-1} \frac{12.69}{30.12} = 22.8^\circ$ Direction : 22.85° below positive x – axis
TO	

NO.	ANSWER SCHEME
	Car : $F - T = m_1 a \rightarrow (1)$
(a)(i)	Wood : $T - f = m_2 a \rightarrow (2)$ $N_2 = m_2 g = (450)(9.81) = 4414.50 \text{ N}$ $f = \mu N = (0.15)(4414.50) = 662.18 \text{ N}$ $a = \frac{(3.8 \times 10^3) - (662.18)}{(1150 + 450)} = 1.96 \text{ m s}^{-2}$
	$F - T = m_1 a$
(a)(ii)	$T = (3.8 \times 10^3) - (1150)(1.96)$ $= 1546 \text{ N}$
(b)(i)	 All 4 correct – 3 Any 3 correct – 2 Any 2 correct – 1 Oth
	$\sum F_y = ma = 0$
(b)(ii)	$N + F \sin \theta + (-W) = 0$ $N = (50)(9.81) - (300) \sin 30^\circ$ $N = 340.5 \text{ N}$

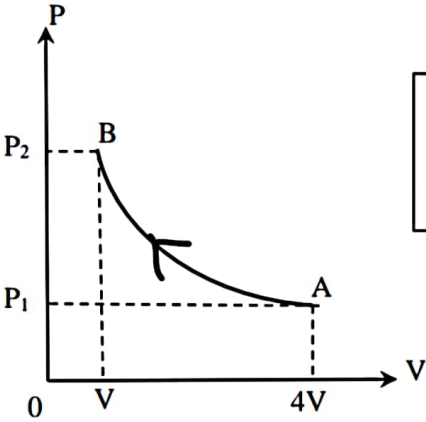
NO.	ANSWER SCHEME
5(a)(i)	$2 \text{ rps} = 2 \text{ Hz}$ $T = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ s}$
5(a)(ii)	$\omega = 2\pi f = 2\pi(2) = 4\pi = 12.57 \text{ rad s}^{-1}$ $v = r\omega$ $= (100 \times 10^{-2})(12.57) = 12.57 \text{ m s}^{-1}$
5(b)(i)	$\theta = 1 \text{ lap} = 2\pi \text{ rad}$ $T = 12 \text{ s}$ $\omega = \frac{2\pi}{T} = \frac{2\pi}{12} = 0.52 \text{ rad s}^{-1}$
5(b)(ii)	$F_c = ma_c = mr\omega^2$ $= (900)(50.0)(0.52)^2$ $= 12168 \text{ N}$
5(b)(iii)	Static friction

NO.	ANSWER SCHEME
6(a)(i)	$v = r\omega \rightarrow \omega = \frac{17}{(33 \times 10^{-2})}$ $= 51.52 \text{ rad s}^{-1}$
6(a)(ii)	$a_t = r\alpha \rightarrow \alpha = \frac{1.12}{(33 \times 10^{-2})}$ $\alpha = 3.39 \text{ rad s}^{-2}$
6(a)(iii)	$\theta = \omega_o t + \frac{1}{2} \alpha t^2$ $= (50)(0.65) + \frac{1}{2} (3.39)(0.65)^2$ $= 33.22 \text{ rad}$
6(a)(iv)	$\text{No. of revolution} = \frac{\theta}{2\pi}$ $= \frac{33.22}{2\pi}$ $= 5.29 \text{ rev}$
6(b)	$\sum \tau = F_1 r \sin \theta + F_2 r \sin \theta$ $= (10)(1) \sin 60^\circ + [-(10)(2) \sin 60^\circ]$ $= -8.66 \text{ N m}$



NO.	ANSWER SCHEME
7(a)(i)	
7(a)(ii)	$\frac{dT}{L} = \frac{(50 - 150)}{(50 \times 10^{-2})}$ $= -200 \text{ }^{\circ}\text{C m}^{-1} \text{ (negative is a must)}$
7(a)(iii)	$A = (3.0) \times (10^{-2})^2 = 3.0 \times 10^{-4} \text{ m}^2$ $\frac{dQ}{dt} = -kA \frac{dT}{L}$ $= -(210)(3.0 \times 10^{-4})(-200)$ $= 12.6 \text{ W}$
7(b)	$M_{\text{nitrogen}} = 28 \times 10^{-3} \text{ kg mol}^{-1}$ $v_{1rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(273.15 + 27)}{(28 \times 10^{-3})}} = 516.95 \text{ m s}^{-1}$ $v_{2rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(273.15 + 352)}{(28 \times 10^{-3})}} = 746.06 \text{ m s}^{-1}$ $\Delta v_{rms} = v_{2rms} - v_{1rms}$ $= (746.06 - 516.95) = 229.11 \text{ m s}^{-1}$



NO.	ANSWER SCHEME
7(c)(i)	 Label axes – 1 mark Shape – 1 mark
7(c)(ii)	Isothermal process = $\Delta U = 0$ Work done on the gas = $-4.5 \times 10^3 \text{ J}$ $\Delta U = Q - W$ $0 = Q - (4.0 \times 10^3)$ $Q = 4.0 \times 10^3 \text{ J}$ $+Q = \text{heat absorbed}$
TO1	